SCIENCE OLYMPIAD FOUNDATION

9th

IMO

LEVEL - 2

Year 2015-16

1. Which of the following sign will replace $*$ in $2 \times 6 - 12 \div 4 * 2 = 5 * 6$?

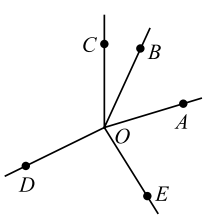
   A. $+$
   B. $-$
   C. $\times$
   D. $\div$

2. Which of the following options hold?
   Statement - 1 : 24 is divisible by 4 and the factors of 4, i.e. 0, 1, 2, 4, 8 also divide 24.

   Statement - 2 : If a number is divisible by another number, then it is divisible by each of the multiple of that number.

   A. Both Statement-1 and Statement-2 are true.
   B. Statement-1 is true and Statement-2 is false.
   C. Statement-1 is false but Statement-2 is true.
   D. Both statement-1 and Statement-2 are false.

3. In the given figure, $\angle AOB = 2\angle BOC$, $\angle AOE = 3\angle BOC$, $\angle EOD = 4\angle BOC$, $\angle DOC = 5\angle BOC$. Find $2\angle AOB + 3\angle BOC$.



   A. 122°
   B. 124°
   C. 168°
   D. 180°

4. Find the value of the expression.

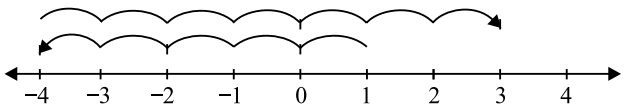
   $-1152 + (-118) - (-32) - (13) - (-56) - (-2183) + 5$

   A. –993
   B. –3137
   C. 993
   D. 389

5. The H.C.F. and L.C.M of $2^3 \times 3^2 \times 5 \times 11$, $2^4 \times 3^4 \times 5^2 \times 7$ and $2^5 \times 3^3 \times 5^3 \times 7^2 \times 11$ respectively are _____.

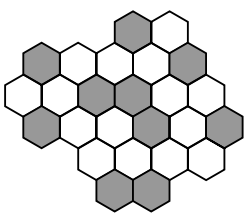
   A. $2^3 \times 3^2 \times 5$; $2^2 \times 3^3 \times 5^2 \times 7$
   B. $2^5 \times 3^4 \times 5^3$, $2^3 \times 3^3 \times 5^3 \times 7^2 \times 11$
   C. $2^3 \times 3^2 \times 5$, $2^5 \times 3^4 \times 5^3 \times 7^2 \times 11$
   D. $2^5 \times 3^4 \times 5^3 \times 7^2 \times 11$, $2^3 \times 5^3 \times 7^2 \times 11$

6. Which of the following number sentence is represented on the number line shown below?



   A. $-2 + (-3) = 1$
   B. $1 + (-2) = -2$
   C. $1 - (+2) = -1$
   D. None of these

7. What percentage of the given figure is shaded?

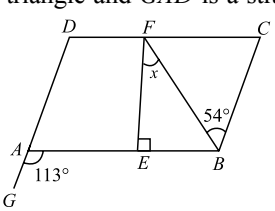


   A. 38%
   B. 60%
   C. 40%
   D. 50%

8. The value of $10^{-3} \times \left(\frac{96.54 - 89.63}{96.54 + 89.63} \div \frac{965.4 - 896.3}{9.654 + 8.963} \right)$ is equal to _____.

   A. 10^{-4}
   B. 10^{-1}
   C. 10
   D. 10^{-5}

9. In the given figure, $ABCD$ is a parallelogram, BFE is a right-angled triangle and GAD is a straight line. Find x.



   A. 30°
   B. 31°
   C. 32°
   D. 33°

10. If $m = \frac{4pq}{p+q}$, then the value of $\frac{m+2p}{m-2p} + \frac{m+2q}{m-2q} =$ _____.

    A. 2
    B. 4
    C. $\frac{2mpq}{(p+q)}$
    D. None of these

11. The parking fee of a certain car park is as follows:

For the 1st hour	₹ 2.50
For every additional half an hour or part thereof	₹ 0.90

If Kartikay parks his car in the car park for 5 hrs 50 mins, how much must he pay?

A. ₹ 8.75
B. ₹ 9.90
C. ₹ 11.50
D. ₹ 17

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12. Aman, Rahul, Raj and Nikhil each took the same spelling test.

- Aman spelled 7/10 of the words correctly.
- Rahul spelled 3/4 of the words correctly.
- Raj spelled 4/5 of the words correctly.
- Nikhil spelled 2/3 of the words correctly.

Who spelled the least number of words correctly and arrange their fraction in ascending order?

A. Aman, $\frac{7}{10} < \frac{3}{4} < \frac{4}{5} < \frac{2}{3}$
B. Rahul, $\frac{3}{4} < \frac{2}{3} < \frac{4}{5} < \frac{7}{10}$
C. Raj, $\frac{2}{3} < \frac{3}{4} < \frac{7}{10} < \frac{4}{5}$
D. Nikhil, $\frac{2}{3} < \frac{7}{10} < \frac{3}{4} < \frac{4}{5}$

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13. If $\frac{a}{3} = \frac{b}{4} = \frac{c}{7}$, then $\left(\frac{a+b+c}{c}\right)^4$ is equal to ____ .

A. 17
B. 16
C. $\frac{1}{16}$
D. 2

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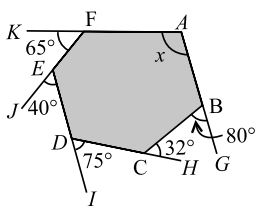
14. Latika has to type a 5-pages document. Each page contains an average of 840 words. How long will she take to type the whole document if she types at the rate of 30 words per minute?

A. $1\frac{1}{12}$ hrs
B. $2\frac{1}{3}$ hrs
C. $1\frac{1}{2}$ hrs
D. $2\frac{3}{4}$ hrs

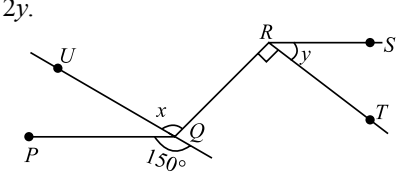
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15. Find the value of $\frac{2x-9}{5}$.

A. 92
B. 22
C. 112
D. 43



16. If $RS \parallel PQ$ and $RT \parallel UQ$, then find the value of $8x - 2y$.



A. 660
B. 30
C. 650
D. 100

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17. A boy multiplied 987 by an integer and obtained 559981 as his answer. If in the answer both 9's are wrong and the other digits are correct, then the correct answer would be _____.

A. 553681
B. 555181
C. 555681
D. 556581

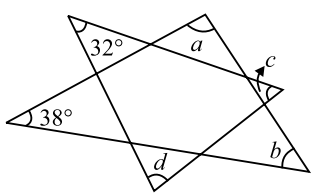
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18. The value of $\frac{3}{2}$ of $\left(\frac{3\frac{1}{2} \div 1\frac{1}{2} \text{ of } 5\frac{1}{2}}{2\frac{1}{2} + 1\frac{1}{2} \text{ of } 3\frac{1}{2}}\right)$ is _____.

A. $\frac{56}{99}$
B. $\frac{56}{1023}$
C. $\frac{28}{341}$
D. $\frac{182}{31}$

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19. Find the sum of $\angle a$, $\angle b$, $\angle c$ and $\angle d$.



A. 280°
B. 290°
C. 360°
D. 240°

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20. Read the below statement carefully and complete it by selecting the correct option.
Parth purchased a machine for ₹ 80,000 and spent ₹ 5000 on repairs and ₹ 1000 on transport and sold at ₹ 1,07,500. He earns _____ on this transaction.

A. Profit of ₹ 32000
B. Profit of ₹ 25900
C. Profit of ₹ 21500
D. Profit of ₹ 19500

21. Shalini borrowed ₹ 800 which amounts to ₹ 1164 in $3\frac{1}{2}$ yrs. What would be the amount in $5\frac{1}{2}$ yrs?

A. ₹ 1800
B. ₹ 364
C. ₹ 1372
D. ₹ 850

22. Two numbers are in the ratio 5 : 3. If 10 is subtracted from both the numbers, their ratio changes to 5 : 7. The sum of numbers initially was _____.

A. 8
B. 15
C. 13
D. 12

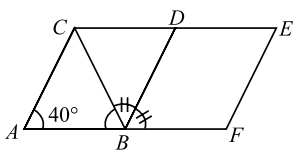
23. Anshu started a business investing ₹ 2,25,000. After 6 months Divya joined her with ₹ 3,25,000. Out of a annual total profit of ₹ 19995, Divya's share is _____ .

A. ₹ 15400
B. ₹ 17200
C. ₹ 19600
D. None of these

24. If three quarters kg of sugar costs 90 paise, then how many paise will 200 gm cost?

A. 48 paise
B. 24 paise
C. 56 paise
D. 72 paise

25. In the given figure, ABF is a straight line. If $AB = AC$, then $\angle ACB - \angle DBF$ is _____.



A. 30°
B. 20°
C. 10°
D. 15°

26. If books bought at price ranging from ₹ 200 to ₹ 350 are sold at price ranging from ₹ 300 to ₹ 425, what is the greatest possible profit that might be made in selling eight books ?

A. ₹ 400
B. ₹ 600
C. ₹ 1800
D. ₹ 2000

27. Which of the following is true for the value of $4xy(x - y) - 6x^2(y - y^2) - 3y^2(2x^2 - x) + 2xy(x - y)$.

A. –195 for $x = 0$ & $y = 1$
B. 2535 for $x = 1$ & $y = 0$
C. –2535 for $x = 5$ & $y = 13$
D. 7605 for $x = 13$ & $y = 5$

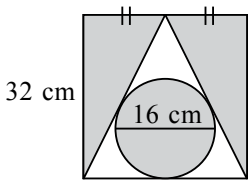
28. A boy was asked to write the value of $(3)^5 \times (9)^2$. He wrote it as 3592. The difference between the obtained value and the actual value is _____.

A. 0
B. 16091
C. 3592
D. 4560

29. The value of $\frac{4x-3}{2}$, if $\left(\frac{8}{5}\right)^3 \times \left(\frac{8}{5}\right)^{-6} = \left(\frac{5}{8}\right)^{1-2x}$ is _____.

A. 0
B. $\frac{5}{2}$
C. $\frac{-1}{5}$
D. $\frac{-7}{2}$

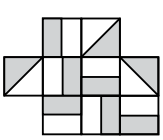
30. The given figure shows a triangle and a circle enclosed in a square. Find the area of the shaded parts. (Take $\pi = 3.14$)

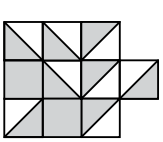


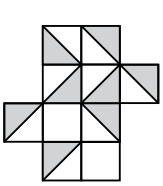
A. 200.96 cm^2
B. 712.96 cm^2
C. 512 cm^2
D. 818.42 cm^2

31. Which of the following is correct with respect to shaded part of a figure?

A.  $= \frac{3}{12}$

B.  $= \frac{4}{9}$

C.  $= \frac{5}{10}$

D.  $= \frac{7}{10}$

32. A circle of radius 2 cm is cut out from a square piece of an aluminium sheet of side 6 cm. What is the area of the left over aluminium sheet? (Take $\pi = 3.14$)

A. 78.5 cm^2
B. 23.44 cm^2
C. 87.92 cm^2
D. 26.54 cm^2

33. The average cost of a shirt, skirt, blouse and a pair of pants is ₹ 73.50. If the average cost of the shirt, skirt and blouse is ₹ 72, find the cost of the pair of pants.

A. ₹ 294
B. ₹ 216
C. ₹ 78
D. ₹ 511

34. Mohit spent 45% of his salary on a laptop. If Mohit had ₹ 1980 left from his salary after buying the laptop, how much was his salary?

A. ₹ 2420
B. ₹ 3069
C. ₹ 2871
D. ₹ 3600

35. Find the range, mean, median and mode respectively of the given data :

75, 200, 98, 75, 100, 98, 75, 96, 180, 270, 102, 94, 100, 610, 200, 75, 80

A. 355, 160, 100, 100
B. 535, 148.70, 98, 75
C. 535, 132.2, 97, 200
D. 5, 150.29, 99, 75

36. Two unbiased dice are tossed. What is the probability of getting a product divisible by 3?

A. $\frac{11}{36}$
B. $\frac{7}{18}$
C. $\frac{5}{9}$
D. $\frac{15}{36}$

37. Find the probability of drawing a face card from a deck of 52 cards.

A. $\frac{1}{2}$
B. $\frac{3}{13}$
C. $\frac{2}{3}$
D. $\frac{1}{4}$

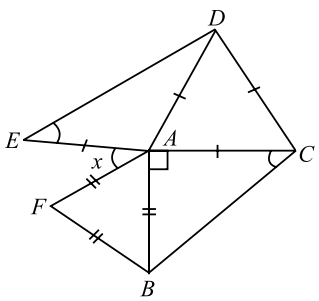
38. The given table shows the charges for parking a car in a mall.

Time	Cost
8 a.m. - 5 p.m.	80 paise per half an hour
After 5 p.m.	₹ 2.15

Priya parked her car from 11.30 a.m. to 6.15 p.m. If she gave a ₹ 100 note to pay parking charges, then how much did she get back after payment?

A. ₹ 80.30
B. ₹ 80.95
C. ₹ 85.85
D. ₹ 89.05

39. In the given figure (not drawn to scale), ADE is an isosceles triangle, BCA is a right-angled triangle, FBA and ACD are equilateral triangles. If $\angle ADE = 45°$ and $\angle ABC = 52°$, then find x.



A. 38°
B. 40°
C. 60°
D. 44°

40. If one angle is the average of the other two angles and the difference between the greatest and least angles is 60°, then the angles of the triangle are in the ratio_____ .

A. 1 : 2 : 3
B. 1 : 4 : 13
C. 1 : 2 : 5
D. 5 : 7 : 11

41. If $6(3x + 2) - 5(6x - 1) - 12x = 6(x - 3) - 5(7x - 6)$, then find the value of $\frac{2x-3}{4x-3}$.

A. 2/7
B. –1
C. 5/7
D. –5/7

42. If 20% of $b = a$, then a% of 20 is the same as _____.

A. 4% of b
B. 5% of b
C. 20% of b
D. None of these

43. Naman bake cakes in a microwave. The microwave takes 10 minutes to warm up before the first tray of cake is baked. Each cake takes 20 minutes to bake. Which of the following tables show the relationship between the number of cakes Naman bakes and the total microwave time?

A.

Number of cakes	Total microwave time (seconds)
1	1800
2	2400
3	3000
4	8600

B.

Number of cakes	Total microwave time (seconds)
1	600
2	1800
3	3000
4	4200

C.

Number of cakes	Total microwave time (seconds)
1	1800
2	3000
3	4200
4	5400

D.

Number of cakes	Total microwave time (seconds)
1	1800
2	3600
3	5400
4	7200

44. If p : every fraction is a rational number and q : every rational number is a fraction, then which of the following options hold?

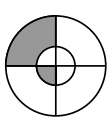
A. p is true and q is false.
B. p is false and q is true.
C. Both p and q are true.
D. Both p and q are false.

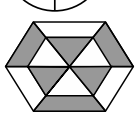
45. A cricket team won 3 matches more than they lost. If a team win gives them 2 points and loss (–1) point, how many matches, in all, they have played if their score is 23?

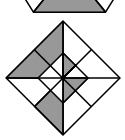
A. 17
B. 20
C. 37
D. 40

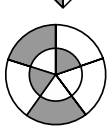
ACHIEVERS SECTION

46. Which of the following shaded fraction represents $4\left[\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right).....\left(1-\frac{1}{10}\right)\right]$?

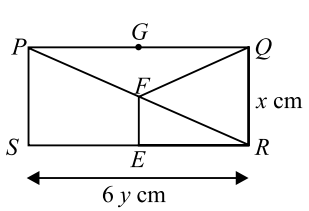
A. 

B. 

C. 

D. 

47. $PQRS$ is a rectangle. E, F and G are the mid points of SR, PR and PQ respectively. The ratio of area ($PSEF$) to area of rectangle $PQRS$ is _____.



A. 9 : 4
B. 3 : 8
C. 5 : 8
D. None of these

48. Given below a question is followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

What is the principal sum?

I. The sum amounts to ₹ 690 in 3 years at S.I.
II. The sum amounts to ₹ 750 in 5 years at S.I.
III. The rate of interest is 5% p.a.

A. Only I and III
B. Only II and III
C. Only I and II
D. Any two of the three

49. Find the value of $\frac{\left(3\frac{2}{3}\right)^2-\left(2\frac{1}{2}\right)^2}{\left(4\frac{3}{4}\right)^2-\left(3\frac{1}{3}\right)^2} \div \frac{3\frac{2}{3}-2\frac{1}{2}}{4\frac{3}{4}-3\frac{1}{3}}$.

A. $\frac{37}{97}$
B. $\frac{74}{97}$
C. $1\frac{23}{74}$
D. None of these

50. A scientist is comparing the weights of the four molecules listed in the given table.

Which one of these molecules is the lightest?

Molecule	Weight (in kilograms)
Salt	9.350×10^{-28}
Pure water	2.879×10^{-26}
Hydrochloric acid	5.832×10^{-22}
Potassium hydroxide	8.976×10^{-24}

A. Salt
B. Pure water
C. Hydrochloric acid
D. Potassium hydroxide

SPACE FOR ROUGH WORK